

Machine Learning — Day 3 Notes

Types of Machine Learning

The three major types are **Supervised Learning**, **Unsupervised Learning**, and **Reinforcement Learning**.

1. Supervised Learning

Definition: The model learns from **data with known answers (labels)**.

Example: A hospital has past patient records showing whether each patient needed another appointment.

Flow: Past Data + Known Answers → Training → Model → New Patient Data → Prediction

Remember: Supervised Learning = **Learning with answers**.

2. Unsupervised Learning

Definition: The data has **no labels**, and the system discovers patterns or groups by itself.

Example: A news app uses users' reading habits to discover different types of readers without predefined groups.

Flow: User Data → Find Patterns → Discover Groups

Remember: Unsupervised Learning = **Finding patterns without answers**.

3. Reinforcement Learning

Definition: An **agent learns by taking actions and receiving rewards or penalties**.

Example: A delivery drone tries different routes. Reaching the destination quickly gives a reward; an inefficient route gives a penalty.

Flow: Current Situation → Choose Action → Environment → Reward / Penalty → Learn → Try Again

Remember: Reinforcement Learning = **Learning through actions and rewards**.

4. Main Difference

Type	Learns from	Example
Supervised	Data + known answers	Predict patient outcome
Unsupervised	Data without answers	Discover reader groups
Reinforcement	Actions + rewards/penalties	Learn drone routes

5. Easy Memory Trick

Supervised → Teacher gives answers

Unsupervised → Find patterns yourself

Reinforcement → Try, get reward/penalty, learn

6. Day 3 Mental Map

Machine Learning



Supervised — Known labels — Predictions

Unsupervised — No labels — Groups/Patterns

Reinforcement — Rewards — Actions

7. Key Definitions

- **Supervised Learning:** Learning from labeled data.
- **Unsupervised Learning:** Finding patterns in unlabeled data.
- **Reinforcement Learning:** Learning through actions and rewards or penalties.